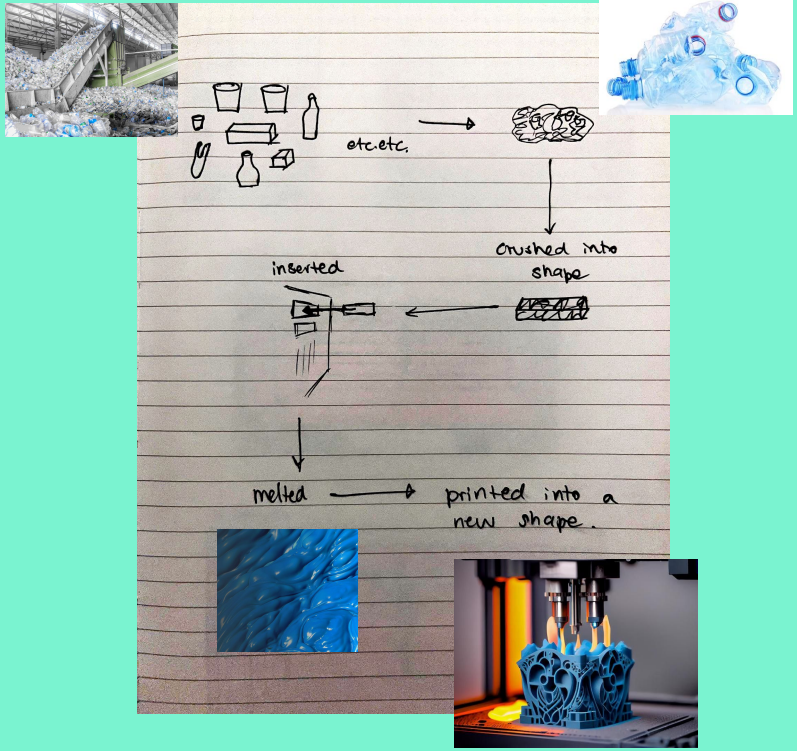
# THE 3D PRINTER

reusing waste plastic to print

# CONCEPT

Our innovative 3D printer design functions as conventional 3D printer but repurposes single use plastics for printing; waste plastic is melted in a heat chamber and then it is injected through the nozzle to print designs using the plastic. This is transformative for the environmental conservation of the Earth as plastic waste in factories, homes and businesses will be reduced. The project leverages the vast quantities of discarded plastics (that make up between 80-90% of all plastic waste globally). As a company, SWS, is innovating for the sake of environment.




SWS

# DESIGN

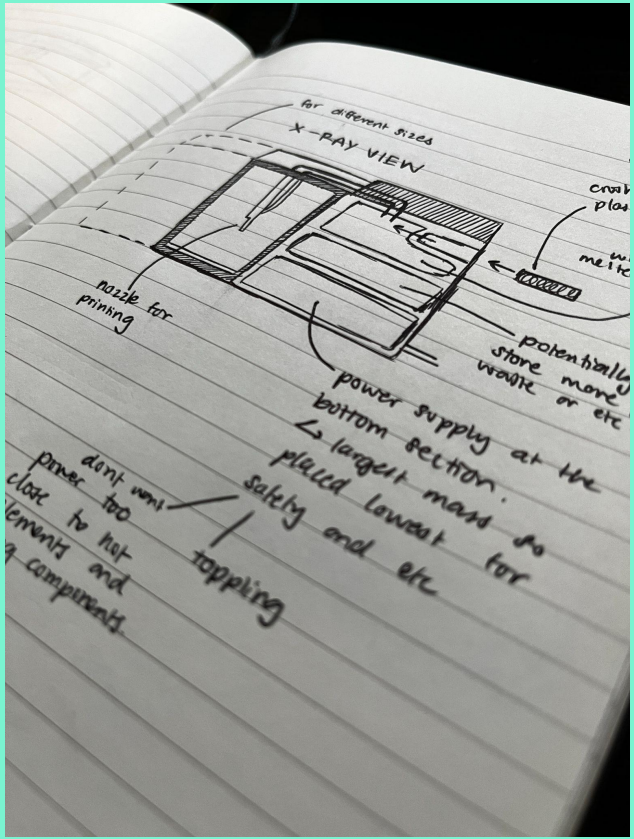
The printing component of the 3D Printer remains to an extent similar to conventional ones. The design however needed to be rethought to move the power source lower down in the machine to lower the centre of gravity for safety against toppling while also keeping the plastic-melting mechanism segregated from the power unit. This was a design choice aimed to optimise safety.

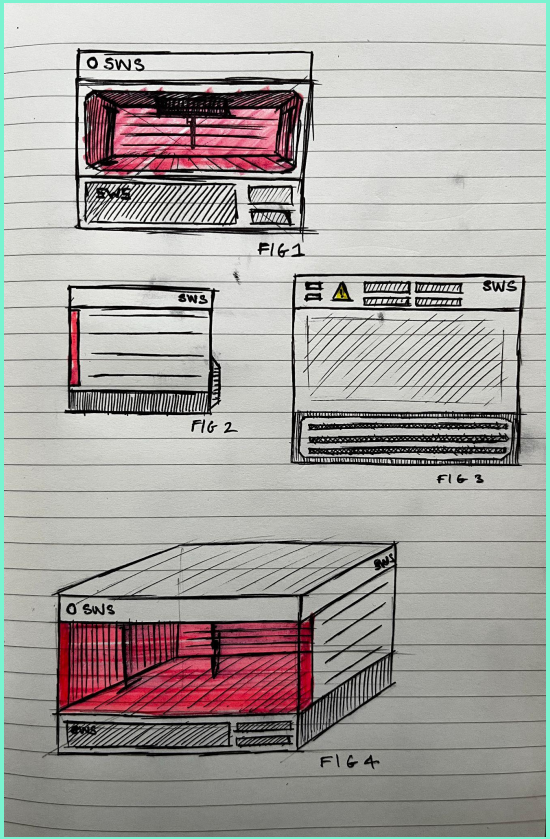
All of the forms of plastic melt under 300*C so there is not immense heat in the machine (safety hazard reduced as well as cost effectiveness guaranteed with not much energy required. This means that the melting will be possible without substantial heaters that could massively disproportion our product’s price to existing printers.

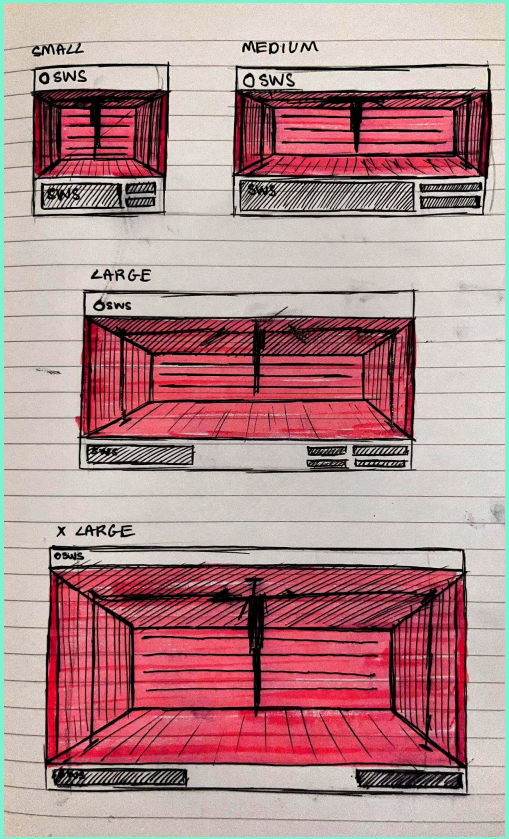
We need a material that is: durable and strong; has high heat and chemical resistance; is affordable and last but certainly not least, eco-friendly and hence we have decided to use:

- We will use HTPLA for non structural components and Nylon for structural components
- Nylon: durable and thus can withstand high stress but can absorb moisture so must be dried.
- HTPLA is biodegradable, has high heat and chemical resistance and and is affordable.

Variations in size are essential for accessing all the possible markets; homes, businesses, industrial corporations. This is because we believe our design can have such an environmental impact that it needs to be widely available.


for different sizes
X-RAY VIEW
nozzle for printing
power supply at the bottom section.
largest mass so placed lowest for safety and etc
dont want power too close to hot elements and components
topping


SWS
FIG 1
FIG 2
FIG 3
FIG 4


SMALL
MEDIUM
LARGE
X LARGE
SWS

# PURPOSE

As a company we realised that there is so much more to recycling than people have realised. Recycling is essential to reducing plastic waste but, unlike our repurposing solution, massive amounts of processing and collection is needed. Here, anyone from individuals to businesses to industrial corporations can make use of our 3D Printing technology.

At home, people can download files for essentials they would otherwise buy and have it printed for them using the waste from things they've already bought. There's no need for delivery of another item (no delivery van, no extra packaging).

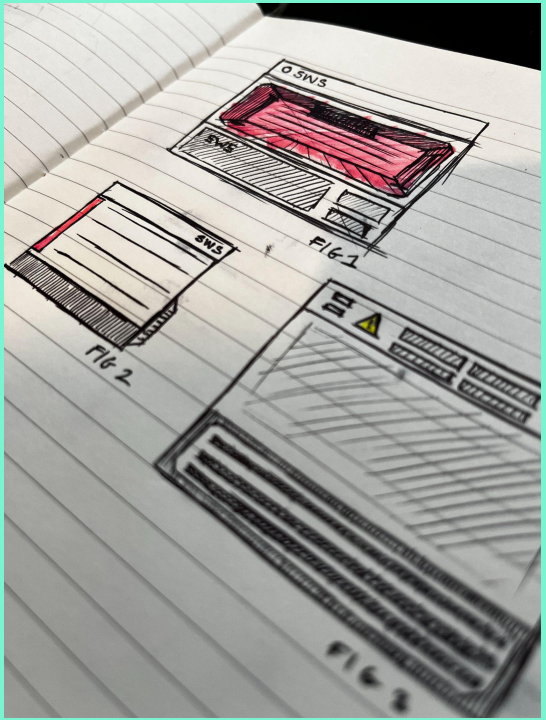
In industrial factories, more can be made without the need for more resources, meaning more profit for businesses without a negative impact to the environment due to waste.

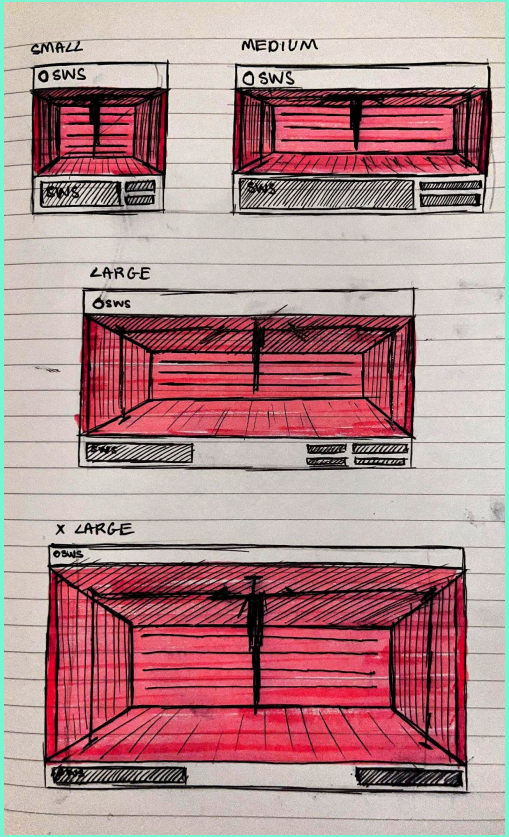
Though our product design is centred around environmental conservation, we see that there are a plethora of benefits and purposes to our 3D printing technology, ensuring there is at least one aspect of our product design that appeals to our vast and widespread target audience.

# APPLICATION

We intend to revolutionise the 3D Printing industry with our idea. Our aim is to divert the trajectory of this industry towards helping the environment with the incentive of reduced waste and improved profits.

To access all sectors that 3D Printing has appealed to, we have chosen as a team to have a variation in sizes of the products.



# COMPETITORS

**Competitors:**

Stratasys Ltd
3D Systems Corps
Proto Labs

Why choose us?

The answer is simple; the concept and application of 3D printers is relatively new; the first 3D printer was invented in 1984 and officially released in 1987. This posits the notion that there is a lack of innovation and originality in the field of 3D printing. We aim to resolve this issue whilst simultaneously playing an imperative role in the global effort of preserving our fragile environmental landscape. Finding an environmentally friendly product that tackles the problem of plastic pollution is at the core of our product design and has become our unique selling point. Naturally, we make sure that we are committed to delivering a product that only matches but exceeds the performance of our competitors.
We penetrate an overlooked yet vital gap in the 3D printing market, the lack of resolve to environmental damage - our design is one that will undoubtedly contribute to the reduction of single plastic waste worldwide.










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